



OLLSCOIL NA GAILLIMHE
UNIVERSITY OF GALWAY

Semester 2 Examinations 2022/2023

Course Instance Codes	3BA1, 4BA161, 4BME1, 4BMU1, 1OA2, 1OA3, 4BCT1, 4BMS2, 4BS2
Exams	3rd and 4th year (Arts and Science), Study Abroad / International Exchange
Module Code	CS402
Module	Cryptography
External Examiner	Prof. C. Roney-Dougal
Internal Examiners	Prof. G. Pfeiffer *Dr K. Jennings
<u>Instructions:</u>	Answer all four questions. Each question carries 25 marks. All workings must be shown.
Duration	2 hours
No. of Pages	3 pages, including this one
Discipline	Mathematics
<u>Requirements:</u>	No special requirements
Release to Library	Yes
Release in Exam Venue	Yes
MCQ	No
Statistical / Log Tables	No
Non-programmable Calculator	Yes
Mathematical Tables	No
Graph paper	No

The following table may be helpful throughout Q1:

A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P
0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Q	R	S	T	U	V	W	X	Y	Z	␣					
16	17	18	19	20	21	22	23	24	25	26					

Q1. (a) The ciphertext string

RFCTEKTWEXFREQTQ

was produced using a shift cipher over the 27-letter alphabet

$$A = 0, B = 1, \dots, Z = 25, \text{␣} = 26,$$

where the final character ␣ indicates a space.

By conducting a frequency analysis of a larger sample of ciphertexts, it has been found that the plaintext symbol ␣ corresponds to the ciphertext symbol E. Determine the first word of the plaintext string corresponding to the ciphertext string given above. [5 marks]

(b) Identify $\mathbf{Z}_{27} = \{0, \dots, 25, 26\}$ with $\{A, \dots, Z, \text{␣}\}$ as in part (a). The ciphertext string **GM␣L** was obtained using a 2-dimensional Hill cipher over $\mathbf{Z}_{27}$ with key

$$\begin{bmatrix} 5 & 1 \\ 1 & 1 \end{bmatrix}.$$

Find the corresponding plaintext. [10 marks]

(c) Describe what is meant by a *brute force attack* and describe a scenario where this is feasible. [5 marks]

(d) An affine cipher (of dimension 1) over $\mathbf{Z}_{360}$ is used to encrypt messages. Find the number of possible keys.

[Please give a number, not a formula.] [5 marks]

Q2. (a) Formally define what is meant by a (symmetric) *cryptosystem*. [5 marks]

(b) Briefly describe what a *perfectly secure* (symmetric) cryptosystem is. Do perfectly secure cryptosystems exist? [5 marks]

(c) State and prove Shannon's Theorem. [15 marks]

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- Q3.** (a) Give an example each of an advantage and a disadvantage of
- symmetric cryptosystems and
 - public-key cryptosystems. [4 marks]
- (b) Describe the key generation, encryption, and decryption steps of the *RSA* cryptosystem. [5 marks]
- (c) Explain why none of the following pairs of integers is a valid RSA key:
- $(n, e) = (187, 15)$.
 - $(n, e) = (289, 13)$.
 - $(n, e) = (385, 77)$. [6 marks]
- (d) A linear feedback shift register $\mathcal{L}$ of length 2 is used to encrypt messages using a stream cipher. It is known that when the cipher is used to encrypt the plaintext string $(x_0, x_1, x_2, x_3) = (0, 0, 1, 1)$, then the corresponding ciphertext string is $(y_0, y_1, y_2, y_3) = (1, 1, 1, 0)$. Find the connection polynomial of $\mathcal{L}$. [5 marks]
- (e) A linear feedback shift register of length 4 with connection polynomial

$$C(X) = 1 + X + X^2 + X^4$$

over $\mathbf{Z}_2$ is used to produce a pseudo-random sequence of binary digits with initial values $s_0 = s_2 = 1$ and $s_1 = s_3 = 0$. Determine the least period of the resulting infinite sequence $s_0, s_1, \dots$ [5 marks]

- Q4.** (a) Define what it means for a group to be *cyclic*. State (without proof) Gauss's characterisation of those $n \geq 1$ for which the group $\mathbf{Z}_n^\times$ is cyclic. Is $\mathbf{Z}_{169}^\times$ cyclic? Justify your answer. [5 marks]
- (b) Describe the method of *fast exponentiation* for computing $a^e \bmod n$, where $a, e, n \in \mathbf{Z}$ with $e, n \geq 1$. [5 marks]
- (c) Describe the key generation, encryption, and decryption steps of the *ElGamal* cryptosystem. [5 marks]
- (d) Alice and Bob use the cyclic group $\mathbf{Z}_{59}^\times$ and its generator $g = 2$ to perform a Diffie-Hellman key exchange. Alice secretly chooses $a = 3$ and Bob secretly chooses $b = 4$. Determine their shared key (as an element of $\mathbf{Z}_{59}^\times$). [5 marks]
- (e) Determine the set of points E of the elliptic curve $y^2 = x^3 + 2x^2 + 2x$ over $\mathbf{Z}_5$. Identify an element of order 2 of E . [5 marks]